

4

A THEORETICAL FRAMEWORK IN THE ACQUISITION AND TEACHING OF FRICATIVES TO L2 LEARNERS OF SPANISH

A. Raymond Elliott

4.1 Introduction

The purpose of this chapter is to provide an overview of research studies examining the acquisition of **fricatives** by **L2** learners of Spanish and the practical application of these theories to classroom methodologies. This chapter focuses specifically on the Spanish fricatives /θ, s, x/ and allophonic (see **allophone**) variants [ð, z, ɣ, h]. Several helpful techniques are outlined, along with suggestions for the design and execution of pronunciation lessons for teaching Spanish fricatives related to: 1. Phonetic (see **phonetics**) instruction regarding concrete rules of **place** and **manner of articulation** (Elliott 1995b, 1997, 2003; Arteaga 2000; Lord 2005, 2010), 2. Auditory (see **auditory phonetics**) exposure through listening comprehension passages and/or **minimal pair** lists, with student attention focused on sounds over meaning, 3. Teaching sound discrimination through the use of articulatory (see **articulatory phonetics**) diagrams, oral repetition, and identification tasks (Elliott 1995b, 1997, 2003; González-Bueno 1997), and 4. The implementation of online resources to complement pronunciation instruction and learning, such as the *Sounds of Speech* (the University of Iowa) and Praat (Boersma and Weenink 2015). Practical recommendations are offered to teachers of all levels for incorporating research-based pronunciation instruction easily into their everyday lesson plans. Although the majority of the research articles discussed here focus on native English speakers learning Spanish, recommendations can be modified and applied to learners of any **L1** background.

4.2 Fricative consonants

Ladefoged and Maddieson (1996, 137) characterize fricatives as sounds in which a turbulent airstream is produced within the vocal tract and is held for a relatively

long period of time.¹ They note that with many fricatives, particularly with **sibilant** varieties (e.g., [s] and [z]), there is a well-defined shape of the vocal tract and a narrowing effect through which air is guided and friction is produced. Although they state that fricatives are “held for a noticeable amount of time” (137) relative to other sounds, they do not indicate precisely how long that may be nor do they provide any indication of appropriate average durations of these sounds. Martínez Celdrán (2004) states that, from an articulatory point of view, fricatives consist of incomplete contact in the vocal tract from which the airstream exits in a constricted fashion and flows against articulators, producing a hissing sound. He argues that fricatives have three main characteristics: narrowing of the vocal tract, friction, and noise.

Historically, the terms **spirant**, **strident**, and **continuant** have been used synonymously with *fricative*; however, Martinet states that fricatives consist of “articulación firme, cerrada, con frotamiento del aire donde se oye la fricción,” while spirants have a more relaxed, open articulation lacking in friction or airstream turbulence (1980, 436). What Martinet proposes is an overall class called *continnants* that consists of both fricatives and spirants. Although the terms *strident* and *continuant* appear to be less commonly used today, not all linguists agree as to what constitutes a spirant. Some view this term as synonymous with fricative (Martínez Celdrán 2004), while others, such as Bowen and Stockwell (1960), Thomas, Bouquiaux, and Cloarec-Heiss (1986), Zampini (1994), and Alvord and Christiansen (2012) define it similarly to Martinet (1980). In this chapter, the term *fricative* is used to refer to those sounds in which a high degree of friction is produced rather than *spirant* or *continuant*, which more aptly describe the pronunciation of the Spanish **approximants** (e.g., [β, ð, ʝ]), which are sounds that have been described as fricatives by some scholars (see Chapter 3 of this volume).

Table 4.1 lists Spanish fricative **phonemes** and **allophones** in both standard Latin American and Castilian dialects, and compares them to those of English. Standard Latin American Spanish has a total of three fricatives, /f, s, x/,² while Peninsular Spanish has a **voiceless interdental** fricative /θ/, pronounced similarly to <th> in the English word *think*. In Peninsular Spanish, /θ/ corresponds with the **graphemes** <c> (before <i, e>) and <z>. While words such as *sima* and *cima* are **homophonous** in Latin American dialects of Spanish (i.e., ['si.ma]), they form minimal pairs in Spain, ['ʝi.ma] versus ['θi.ma], respectively.

By comparing the fricatives of Spanish and English, specific areas of potential difficulty for non-native speakers learning Spanish become apparent. The Spanish **consonants** [z, h] are allophones of /s, x/, respectively, and do not surface

TABLE 4.1 Spanish and English fricatives (*denotes allophones)

Language	Manner	Labiodental	Interdental	Alveolar	Palatal	Velar	Uvular	Glottal		
Spanish:	Fricative	f	θ	ð*	s	z*	x	χ*	h*	
English:	Fricative	f	v	θ	ð	s	z	ʃ	ʒ	h

phonemically in Spanish as they do in English. In addition, Spanish has fricatives not found in English (e.g., [x, χ]). Likewise, English has three additional fricative consonants /v, ʃ, ʒ/, appearing in words such as *vivid* ['vi.vəd], *shirt* [ʃəɹt], and *measure* ['me.ʒə], that are not found in standard Spanish.

Spelling differences between the L1 and L2 may also present potential pitfalls for non-native Spanish speakers; for example, <v> in many languages is articulated as a **voiced labiodental** fricative, but in Spanish, it is a voiced **bilabial** stop /b/ that can be articulated as [b] or as a bilabial approximant [β]. Likewise, the English grapheme <h> is pronounced as a voiceless **glottal** fricative, whereas in Spanish, it is mute. Consequently, the grapheme-to-phoneme effect and the issue of **orthography**, including L1 and L2 similarities and differences, have to be considered (see Chapter 11). Spelling differences between the learner's L1 and L2 may both facilitate and hinder the acquisition of L2 pronunciation. In addition, when teaching pronunciation, one must take into account dialectal variants in the pronunciation of specific sounds; for example, [ʃ] and [ʒ] can be found in Argentinian Spanish but do not occur in standard Latin American or Castilian Spanish. Furthermore, **/s/-aspiration** and **/s/-deletion** dialects (i.e., dialects that weaken /s/) should also be considered when teaching Spanish fricatives to L2 learners. Several areas of potential difficulty in the acquisition and production of Spanish fricatives are discussed below.

4.3 What sounds to teach?

Not all fricatives contribute equally to foreign sounding accents when speaking Spanish. Therefore, prior to designing course curricula, we must first determine which fricatives should be taught. Elliott administered a survey to several Spanish instructors, professors, and graduate teaching assistants in which they were asked to list Spanish sounds they felt contributed most to a foreign accent, which is defined as any oral linguistic output that a native Spanish speaker would deem as sounding unnatural or non-native (1997, 98). With specific regard to fricatives,³ teachers listed the voicing of /s/ (see **/s/-voicing**) before voiced consonants as problematic for non-native learners of Spanish, while none of those surveyed deemed /f/ as problematic. A handful of instructors noted that native English speakers have difficulty pronouncing [χ], the voiceless **uvular** fricative in Castilian Spanish, and others stated that students may overgeneralize their pronunciation of the voiceless interdental fricative /θ/ to **phonetic contexts** where it should not occur, specifically with regard to ways in which this sound is written in Spanish. In addition, several instructors noted that the Spanish grapheme <h> is frequently pronounced as a voiceless glottal fricative, as in English and other languages, whereas in all dialects of Spanish, this grapheme has no acoustic (see **acoustic phonetics**) representation.

Perhaps the answer regarding which fricatives should be taught can be found through contrastive phonological (see **phonology**) analysis. **Contrastive Analysis** (CA) was one of the best-known and most widely researched theories of

language acquisition from the 1950s through the 1970s (Lado 1957; Wardhaugh 1970). Under CA, it was believed that language learning consisted mainly of habit formation, and that errors in the L2 could be traced back to the learner's L1. Consequently, similarities between the L1 and L2 would be easy to learn, while differences would be more difficult. Stockwell and Bowen (1965) were among the first to propose a hierarchy of phonological difficulty by comparing the sound inventory of English to that of Spanish. The hierarchy, illustrated in Table 4.2, predicts that sounds that are not present in the learner's L1, but are obligatory in the target language (TL), are more difficult to acquire; for example, learners of Spanish may find it more difficult to pronounce a voiced **alveolar** trill (see **rhotic**) in words such as *rascar* and *perro* given its absence in their L1. However, sounds that exist in both languages (e.g., the voiceless labiodental fricative /f/) are easier to pronounce due to positive transfer. Although CA fell short in its ability to predict learner errors in morphology (see **morpheme**), syntax, and semantics, the theory continues to have significant implications for L2 phonological acquisition.

If CA truly has predictive validity for L2 pronunciation, the search for universal phonological differences and similarities among the world's most common languages may provide teachers and curriculum designers with the information they need to determine which fricatives should be taught. Using the UCLA Phonological Segment Inventory Database (UPSID 2016), I calculated percentages of occurrence of fricatives in approximately 460 of the world's most prominent languages. The results are presented in Table 4.3. By examining the data, clustering patterns become apparent, [h, s, f] are found in approximately 40 percent to 62 percent of the world's languages, followed by [x, z, θ], which occur at rates of 13 percent to 22 percent, and finally, the voiceless **velar** fricative [χ], which surfaces in fewer than 10 percent of the world's languages.

Considering the polar extremes presented here, CA would predict that new and unfamiliar sounds (i.e., those sounds that are not as common in the world's languages) are more difficult to learn, while sounds that are more common universally are easier due to positive transfer. It would appear then that sounds that surface less frequently in the world's languages are more **marked** (see **marked form**), and are

TABLE 4.2 Stockwell and Bowen's (1965) hierarchy of phonological difficulty

<i>Native language</i>	<i>Target language</i>	<i>Degree of difficulty</i>
Ø	Obligatory	Difficult ↓ Easy
Ø	Optional	
Optional	Obligatory	
Obligatory	Optional	
Obligatory	Ø	
Optional	Ø	
Optional	Optional	
Obligatory	Obligatory	

TABLE 4.3 Percentage of occurrence of fricatives in the world’s languages, according to the UPSID database

<i>Consonant</i>	<i>Description</i>	<i>UPSID % of occurrence in the world’s languages (most to least common)</i>	
[h]	voiceless glottal fricative	61.8	Easy ↓ Difficult
[s]	voiceless alveolar fricative	45.01	
[f]	voiceless labiodental fricative	40.35	
[x]	voiceless velar fricative	22.84	
[z]	voiced alveolar fricative	15.08	
[θ]	voiceless interdental fricative	13.30	
[χ]	voiceless uvular fricative	9.00	

perceived to contribute more to foreign sounding accents. Considering the data in Table 4.3, CA predicts that /h, s, f/, given their prominence in the world’s languages, should not be as problematic for learners as /θ, z, x/ and [χ]. In other words, the chances are greater that L2 learners of Spanish will already have the sounds /h, s, f/ in their L1, and should be able to transfer these sounds to the L2 with little to no difficulty, provided that these sounds surface phonemically and not phonetically in either the L1 or L2. Research on phonological acquisition lends credence to this claim, and shows that phonetic differences are more marked and more difficult to acquire than are phonemic differences, a finding that has implications for the teaching of L2 pronunciation.

Based on the findings here, and on teachers’ assessment of sounds that are problematic for L2 learners of Spanish, this chapter focuses on fricatives that contribute more to foreign sounding accents, such as /θ, s, x/, and will not include Spanish /f, h/, which tend to be more easily acquired by L2 learners of Spanish, regardless of their L1.⁴ There are two additional reasons for not focusing on /h/ in this chapter. First, although [h] may occur as an allophone of /x/ in some dialects of Spanish spoken in Northern Mexico and Colombia, it may be used as an appropriate substitute for the pronunciation of /x/. Second, although [h] may also occur as an allophone of /s/ in **coda** position in some areas of the Caribbean (e.g., *¿Cómo estás?* as [‘ko.mu.ɐh.‘t̪aɦ]), its distribution is relatively restricted and can be taught as a possible variation in pronunciation, depending upon the dialectal needs of learners.

4.4 Research examining the acquisition of Spanish fricatives

Research on the acquisition of L2 pronunciation falls squarely into three major categories: 1. Exploratory studies designed to identify individual learner characteristics that perhaps explain and account for differences in the acquisition of Spanish pronunciation (e.g., Zampini 1994; Face and Menke 2009; Ringer-Hilfinger 2012; George 2014), 2. Studies that examine the effect of formal instruction on pronunciation (Castino 1992; Elliott 1995a, 1995b, 1997; González-Bueno 1997; Lord

2008; Bajuniemi 2013), and 3. Studies that assess the effect of study abroad and the acquisition of Spanish pronunciation with specific regard to [θ], [χ], /s/-weakening, /s/-aspiration, and /s/-deletion (Geeslin and Gudmestad 2008; Ringer-Hilfinger 2012; George 2014).

While many studies examining the effect of formal instruction on pronunciation have shown that classroom lessons can result in significant improvement in the pronunciation of individual TL sounds, others have shown that formal instruction has either no relationship to pronunciation ability (MacDonald, Yule, and Powers 1994) or that it may even have possible negative effects on pronunciation ability, leading to the overgeneralization of TL sounds (Elliott 1995b). Although the pronunciation of **segments** has been shown to improve as a result of formal phonetic instruction, researchers argue that even after formal pronunciation study, subjects fail to achieve native or native-like pronunciation. With specific regard to Spanish fricatives, Elliott (1997) found that sounds that differ in comparison to English due to positional constraints (e.g., the voicing of /s/ before a voiced consonant) evidenced little to no improvement as a result of formal instruction. In addition, Elliott also reported that the pronunciation accuracy of some sounds declined due to L1 transfer when subject attention was focused on communicating meaning and not on pronunciation. In other words, subjects' L2 pronunciation was effected more by their L1 phonological systems when they were talking about everyday situations (e.g., when describing a classroom or a party) in comparison to their pronunciation ability when they were engaged in word reading or in word and sentence repetition exercises. This finding contradicts both Zampini (1994) and González-Bueno (1995), who found that learners tend to produce some TL sounds better when they focus on communicating meaning as opposed to pronunciation. It would appear then that task formality plays a significant role in the acquisition of L2 pronunciation; however, the ways in which it does so remain to be determined.

Several research studies have examined Spanish fricatives within a study abroad context. While many of these studies focus on the acquisition of [β, ð, γ] (Díaz-Campos 2004; Lord 2010), a handful examined the acquisition, perception, and production of the Spanish fricatives /θ, s, x/. Geeslin and Gudmestad (2008) studied the frequency of usage of /θ/ and /s/-weakening by 130 university L2 learners of Spanish who studied abroad in Spain. Using monologic role plays with Power-Point, they found that only nine out of 130 subjects pronounced the voiceless interdental fricative /θ/, and that just five evidenced /s/-weakening. In addition, subjects scoring in the two lowest proficiency levels never produced either variant. Geeslin and Gudmestad (2008) concluded that study abroad in Spain contributed little to the acquisition of these two sounds. In addition, they found that subjects' desire to fit into the TL culture played a non-significant role in acquiring these sounds. When surveyed, the majority of students indicated they were aware of Spanish /θ/; however, very few mentioned /s/-weakening. Geeslin and Gudmestad (2008) argue that the use or non-use of these two sounds may be affected by individual characteristics such as attitude, desire, and motivation, and that the

decision to not use these sounds was consciously made. They hypothesize that /s/-weakening was probably avoided by their subjects because word-final /s/ is necessary for communicating meaningful information, both in the formation of plural nouns (e.g., *casa* ~ *casas*) and as a person marker in verbs (e.g., *habla-s*, *hablará-s*). The same is not true, however, for Spanish /θ/ insofar as native speakers can understand non-native speakers perfectly well if they use /s/ rather than /θ/.

In another study, Ringer-Hilfinger (2012) examined learners' awareness of the voiceless interdental fricative /θ/ in relation to native-like frequency of usage and attitude toward this sound. She found that students who had previously studied in Spain demonstrated a higher awareness of /θ/ in comparison to pre-study abroad students. In addition, she reported that usage of /θ/ varies according to the dialect students came into contact with before and after living abroad. In other words, if students had been exposed early on to the usage of /θ/, the chances were greater that they would use this sound when subsequently studying abroad. Although Ringer-Hilfinger's findings demonstrate how exposure to TL sounds can lead to greater levels of perception and awareness, this does not guarantee that learners will use these sounds when speaking the TL, especially for sounds that have appropriate target-like substitutes.

Knouse (2012) investigated the production of /θ/ by 15 university students who were studying in Salamanca, Spain for a six-week period, and compared their acquisition of /θ/ to that of ten at-home students enrolled in an Introduction to Spanish Linguistics class in the United States, all of whom had previously studied abroad in Spain. While students living in Spain were exposed to the use of /θ/ on a daily basis, the instructor of the at-home students spoke Castilian Spanish and explicitly instructed them on the usage of /θ/. In spite of having been exposed to /θ/ in forms of input, Knouse found that study abroad students rarely used this sound in **obligatory contexts**, and that at-home learners did not use this sound at all. Although gains made in the pronunciation of /θ/ by study abroad students were statistically significant (due to the at-home learners not using /θ/ at all), Knouse concluded that study abroad contributed little to the acquisition of this phoneme. She explains that the low usage of /θ/ can be attributed to learners' belief that the voiceless interdental fricative is not absolutely necessary when speaking Spanish, and that /s/ is an appropriate substitute for /θ/.

George (2014) examined the acquisition of /θ/ and [χ] by 25 Spanish majors and minors from various universities in the United States during their study abroad in Spain. She found that subjects who had traveled to Spain or had an instructor who spoke with a Castilian accent tended to have better pronunciation of the Castilian /θ/. George also found that learners used /θ/ more in a word list task than they did when reading a passage or when conversing, once again, this indicates that task formality may influence L2 pronunciation. In addition, subjects who used /θ/ tended to overgeneralize its usage to non-obligatory contexts, overlapping this sound with /s/. On the other hand, subjects who had higher proficiency levels in Spanish, more positive attitudes toward Castilian Spanish, and who were less anxious about their L2 pronunciation, had better pronunciation of the Spanish [χ].

Based on the findings reported in the studies above, it is clear that many factors come into play when acquiring an L2 phonological system. We will see that these studies have important implications for the acquisition of Spanish fricatives, and provide teachers and curriculum designers with empirical research they can use when designing theoretically based pronunciation lessons.

4.5 Teaching Spanish fricatives

The following sections address the Spanish fricatives /θ, s, x/, moving from **anterior** to **posterior** places of articulation. Each subsection consists of a phonetic description of the sound, its orthographic representation, problematic areas for L2 learners, and recommended classroom activities.

4.5.1 Pronunciation and orthographic representation of /θ/

The phoneme /θ/ is a voiceless interdental fricative that occurs only in Castilian or standard Peninsular Spanish. It is pronounced by placing the tip of the **tongue** between the upper and lower front teeth or by placing the apex of the tongue “lightly against the back of the upper front teeth” (Dalbor 1980, 85). Hualde (2005, 154) notes that although the **contrast** between /s/ and /θ/ (i.e., **distinction** /s/ ~ /θ/), as in *sien* /sien/ versus *cien* /θien/, is part of standard Peninsular Spanish, it is not found in the Canary Islands or in large areas of Andalusia and Extremadura. In Latin America, /θ/ is not found anywhere.

The phoneme /θ/ is written in Spanish as <c> before the front **vowels** /i, e/, and as <z> before /e, a, o, u/, for example, *encima* [eŋ.'θi.ma] (for details on **assimilation** of /n/ in this context, see Chapter 6) and *zozobra* [θo.'θo.βra]. Some Spanish words, such as *cebra*, may be written with <c> or <z>. Except for words of foreign or scientific origin, such as *zigzag* or *zígoto*, <z> does not usually appear before <i>. When Spanish <z> surfaces before a voiced consonant, it voices to [ð] (e.g., *hazlo* ['að.lo]), similar to <th> in the English word *either* ['i.ðə]. Table 4.4 outlines the distribution of [θ] and [ð] in standard Peninsular Spanish, and provides a formal phonological rule that generates each allophone.

TABLE 4.4 Distribution of /θ/

Allophone	Description	Environment	Examples		
			Word	Phonemic transcription	Phonetic transcription
/θ/ [θ] [ð]	[–voiced] interdental fricative	elsewhere	civil	/θibil/	[θi.ˈβil]
	[+voiced] interdental fricative	before voiced consonant	hazlo	/aðlo/	[ˈað.lo]

The **spectrogram** in Figure 4.1 illustrates differences in the pronunciation of /θ/ before a voiceless consonant, in *hazte* ['aθ.te], and before a voiced consonant, in *diezmo* ['djeð.mo]. The dotted outline in the **waveform** below denotes voicing, or vibration of the vocal cords, while the portion outlined with a solid line denotes a voiceless articulation.

When teaching the voiceless interdental fricative, it bears questioning whether or not teaching /θ/-voicing is warranted. In other words, do native Castilian speakers notice when speakers fail to voice /θ/ in voiced contexts, and if so, does this contribute to the perception of a foreign accent? My overall feeling is they do not. Whether or not a lack of /θ/-voicing before voiced consonants constitutes a perceptually salient pronunciation error for native speakers of Castilian Spanish remains unanswered. If mutual intelligibility is the overarching goal, and if the failure to voice /θ/ does not impede communication or result in seemingly strange pronunciations, then time used for teaching pronunciation could be better spent on sounds that appear to contribute more to foreign-like pronunciation.

Research on the acquisition of /θ/ has shown that, for whatever reason, students fail to incorporate this sound into their L2 phonology. Some studies have shown that many L1 American-English speakers are hesitant to use /θ/ because they associate it with lisping. Although /θ/ exists phonemically in English, L2 learners' hesitation to use this sound in Spanish is due to the fact that the contexts where Spanish /θ/ occurs correlate with the same phonetic contexts where lisping is found in English. In order to combat this perception, George (2014, 107) advocates that

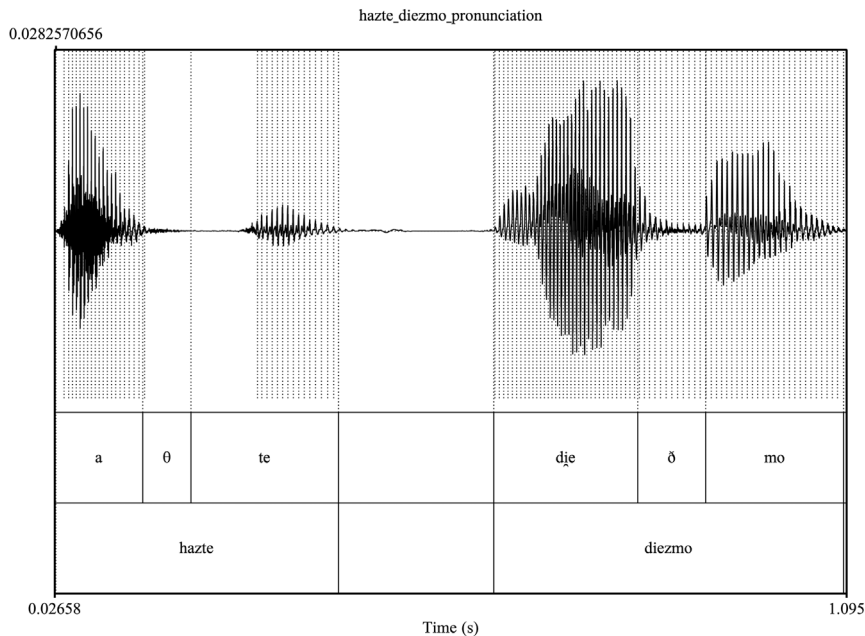


FIGURE 4.1 Differences in voicing of /θ/ before voiceless and voiced consonants.

formal instruction might dispel negative stereotypes. Since both /θ/ and /ð/ exist in English, students would not have to learn to pronounce entirely new sounds, but rather only the appropriate obligatory contexts where these sounds occur; for example, Figure 4.2 enables students to compare Spanish /θ/ and its allophone [ð] to English's /θ/ and /ð/, respectively. By knowing the precise distribution of these sounds in both languages, students may be less hesitant to use them in their appropriate phonetic environments when speaking the TL.

It would appear then that the problem with [θ] and [ð] is not an inability to pronounce these sounds, but rather the orthographic representation of these sounds in Spanish. In other words, many non-native speakers are not quite sure which graphemes represent /θ/ or /s/, especially when /θ/ is represented orthographically as <c> before <e> or <i> in Spanish. On the other hand, the grapheme <z> is always pronounced as [θ] in Spanish, so this association should not be as problematic for L2 learners.

Since some pronunciation problems with [θ] may stem from spelling differences, mastery of the orthographic system and sound–symbol relationships is necessary. Therefore, classroom activities should focus on spelling and the phonetic contexts in which /θ/ undergoes voicing; for example, facial diagrams, such as those in Figure 4.3, can help students learn where to put their tongue when pronouncing [θ] and [ð].

In addition, students and teachers have several Internet resources that can help improve the pronunciation of [θ]. The University of Iowa has an excellent website with videos of native speakers pronouncing TL words, along with animated articulatory diagrams with step-by-step instructions (IURF n.d.). Figures 4.4–4.7, adapted from the contents of this website, outline the articulatory movements involved in the pronunciation of [θ].

Other classroom activities should focus on word and sentence reading or repetition exercises (as described in Elliott 1995b), or those that teach students to carefully monitor their pronunciation, especially if Castilian pronunciation is taught. A sample exercise is provided in Table 4.5.

	English:	Spanish:
Orthographic:	th	z; c + e, i
Phonemic:	/θ/ /ð/	/θ/
Phonetic:	[θ] [ð]	/θ/ /ð/
Examples:	think path they, these those, that	zozobra encima, ceceo diezmo hazle

FIGURE 4.2 Distribution of English /θ/ and /ð/ in comparison to Spanish /θ/.

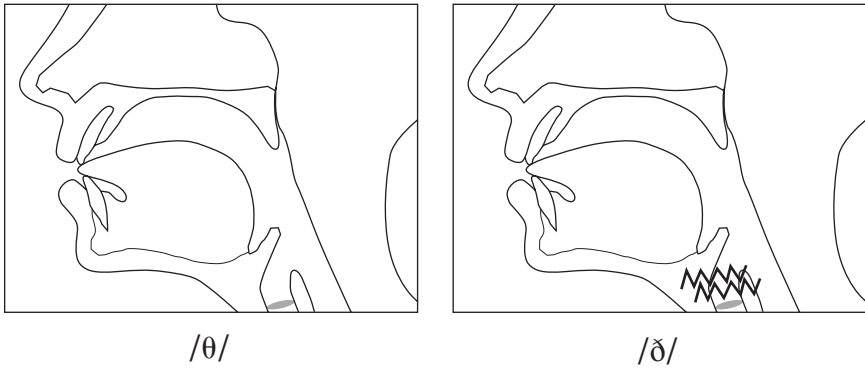
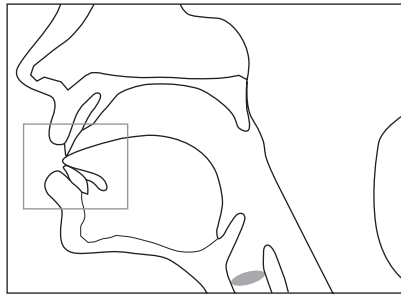


FIGURE 4.3 Facial diagram comparing the pronunciation of [θ] and [ð].

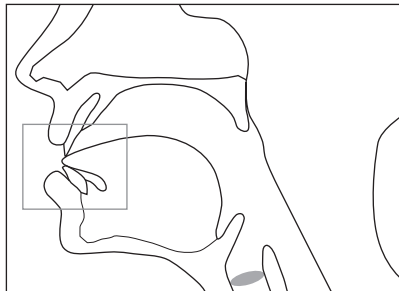
Source: adapted from Hammond (2001, 242).



/θ/ The tip of the tongue becomes narrow and is placed between the edges of the front teeth without completely blocking the flow of air.

FIGURE 4.4 Pronunciation of /θ/, Step 1.

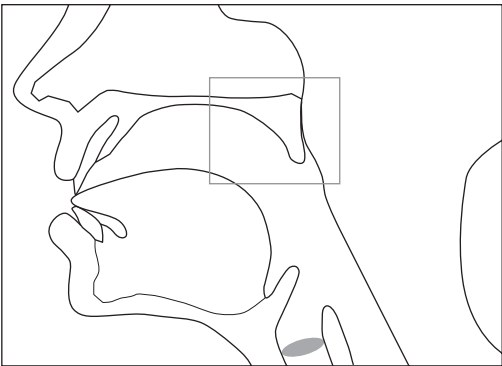
Source adapted from UIRF (n.d.).



/θ/ The sides of the tongue touch the inside portion of the upper molars, preventing air from escaping on each side of the tongue.

FIGURE 4.5 Pronunciation of /θ/, Step 2.

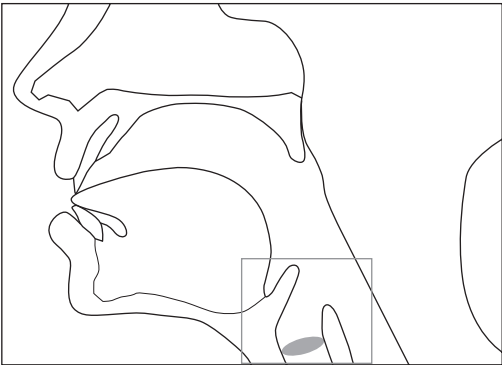
Source: adapted from UIRF (n.d.).



/θ/ The velum remains raised.

FIGURE 4.6 Pronunciation of /θ/, Step 3.

Source: adapted from UIRF (n.d.).



/θ/ The vocal cords do not vibrate.

FIGURE 4.7 Pronunciation of /θ/, Step 4.

Source: adapted from UIRF (n.d.).

TABLE 4.5 Pairs of words illustrating the distribution of /s/ versus /θ/ and [θ] versus [ð]

<i>casa</i>	[ˈka.ɣa]	<i>caza</i>	[ˈka.θa]
<i>seso</i>	[ˈse.ɣo]	<i>ceso</i>	[ˈθe.ɣo]
<i>sima</i>	[ˈsi.ma]	<i>cima</i>	[ˈθi.ma]
<i>has</i>	[aɣ]	<i>haz</i>	[aθ]
<i>rasco</i>	[ˈraɣ.ko]	<i>rasgo</i>	[ˈraɣ.ɣo]
<i>hazte</i>	[ˈaθ.ɣe]	<i>hazlo</i>	[ˈað.lo]

4.5.2 Pronunciation and orthographic representation of /s/ > [s], [z]

The phoneme /s/ is a voiceless alveolar fricative that voices to [z] before voiced consonants, as in *mismo* ['miz.mo]. When pronouncing [s] or [z], the tip of the tongue lightly touches the back of the bottom lower teeth while the blade of the tongue presses against the alveolar ridge (Dalbor 1980, 85). In standard Peninsular Spanish, /s/ is apico-alveolar [ɰ] and is pronounced similarly to <sh> in the English word *shallow*, but with less noise (Dalbor 1980). For [ɰ] and [ʒ], the apex of the tongue approaches the alveolar ridge and is slightly retroflexed (see **retroflex**) (Dalbor 1980, 85).

In Latin America, [s] and [z] are written as <s>, <z> or <c> before [e, i]. In Peninsular Spanish, [ɰ] and [ʒ] are always written as <s> (for information regarding the Peninsular Spanish pronunciation of <z> and <c>, see Section 4.5.1). Table 4.6 outlines the distribution of [s] and [z] in standard Spanish, and provides a formal phonological rule that generates each allophone.

Depending on the geographical area and dialect, Spanish /s/ has several different pronunciations. Schmidt (2014) states that **syllable**-final /s/ frequently undergoes aspiration and may be deleted, thus resulting in other non-sibilant variants of /s/. Although prescriptively this pronunciation may be viewed as non-standard, Hammond (2001) estimates that approximately 50 percent of the world's Spanish-speaking population aspirate or eliminate syllable- and word-final /s/ altogether. Schmidt (2014) argues that it is not uncommon to imagine learners being exposed to /s/-deletion and /s/-weakening, and further notes that in areas where native speakers generally retain /s/, this sound may not voice before a voiced consonant, depending on the dialect.

L1 orthography has also been found to influence learners' pronunciation of Spanish /s/ (Dalbor 1980; Elliott 1997; Teschner 2000). In English, native speakers frequently voice <s> in intervocalic and word-final positions, for example, *president* ['pre.zə.dent] or *girls* [gɜ:lz].⁵ Due to the orthographic distribution of <s> in English, learners may voice /s/ in similar environments in Spanish, as in *rosa* and *mesa* as ['ro.za] and ['me.za] rather than ['ro.sa] and ['me.sa], respectively. To illustrate this point, the Praat image in Figure 4.8 compares two native English-speaking students' pronunciations of *rosa*, the first token is pronounced as ['ro.sa] and the second as ['ro.za]. Both the

TABLE 4.6 Distribution of /s/

Allophone	Description	Environment	Examples		
			Word	Phonemic transcription	Phonetic transcription
/s/ [s] [z]	[-voiced] alveolar fricative	elsewhere	seso	/seso/	['se.so]
	[+voiced] alveolar fricative	before voiced consonant	mismo	/mismo/	['miz.mo]

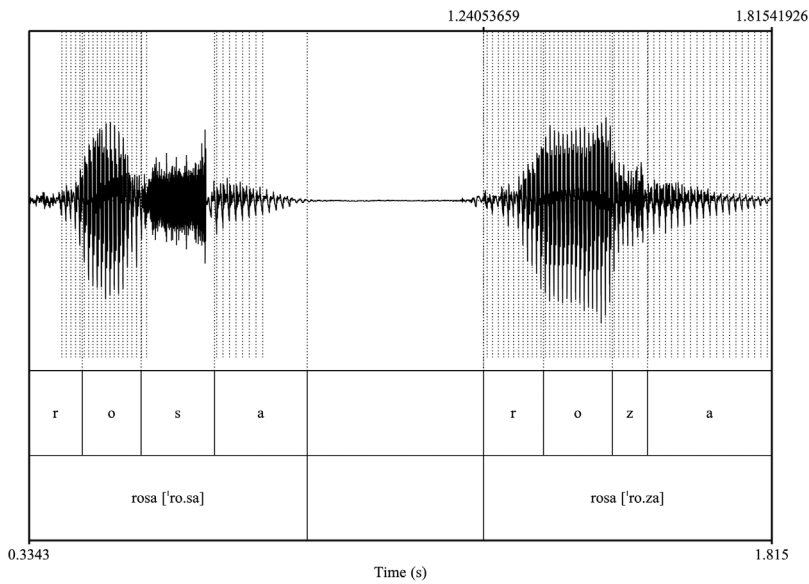


FIGURE 4.8 Waveform and spectrogram of two productions of *rosa*.

waveform and spectrogram highlight differences in the pronunciation of intervocalic /s/. The first is more similar to standard Spanish pronunciation because it is voiceless, while the second is voiced, as evidenced by the pulses (i.e., voicing), which are most likely the result of L1 interference coming from the pronunciation of the English word *rose* [rɒʊz]. As shown in Figure 4.8, Praat is an excellent tool that teachers and students can use for comparing voiced and voiceless articulations.

Does failure to voice /s/ in voicing contexts lead to the perception of foreign accent? McBride (2015) found that native Spanish speakers did not identify failing to voice /s/ as a contributing factor to foreign accent, but did observe English-speakers' tendency to lengthen /s/ when pronouncing this sound.⁶ As such, it would appear that more lengthened pronunciations of /s/ contribute more to foreign accent than does voicing of this sound in inappropriate contexts.

Although Dalbor (1980, 87) states that there are many native Spanish speakers for whom [s] ~ [z] occur in **free variation**, research on Spanish phonology has identified several linguistic factors that condition voicing assimilation of /s/; for example, voicing of /s/ is more frequent before /b, d, g/ and is inhibited before a **liquid** or nasal (see **nasal-stop class**) consonant (Schmidt and Willis 2011). Therefore, applied classroom activities should focus on /s/-voicing in contexts preceding /b, d, g/ and before nasals and liquids (e.g., *esbelto*, *desde*, *hallazgo*, *asma*, and *hazlo*).

Aside from problems with /s/-voicing and lengthening, another pitfall deals with assimilation of /s/ to dentalveolar consonants. In Spanish, when /s/ comes before a **dental** consonant (e.g., [t̪, d̪]), its place of articulation changes from alveolar to dental (Dalbor 1980; Azevedo 2008; Hualde et al. 2010). Again, does this contribute to

foreign sounding accents for native Spanish speakers? In words such as *este* ['es̺.te] or *desde* ['des̺.ðe], the degree of foreign accent perceived by native Spanish speakers appears to result more from pronouncing Spanish [t, d] as alveolar rather than dental than it does from /s/-assimilation in place of articulation. Therefore, failure to assimilate /s/ to dental consonants is most likely imperceptible to linguistically naïve native Spanish speakers. Once students successfully master the correct articulation of /t/ and /d/, pronouncing dentalalveolar [s̺] should come automatically.

Classroom activities for teaching /s/ and its various allophones can consist of facial diagrams, as in Figure 4.9, that compare and contrast standard Latin American Spanish pronunciation of /s/ to that of Peninsular Spanish.

Other activities may consist of repetition exercises, meaningful communicative activities or the use of tongue-twisters (i.e., *trabalenguas*) focusing on pronunciation. Tongue-twisters can be a fun activity for students because they not only help lower learners' affective filters, but also serve to foment student relationships, while at the same time focusing their attention on the pronunciation of segments in running speech. Two of the best Internet tongue-twister sites are *Omniglot: The Encyclopedia of Writing Systems & Languages* (Ager n.d.) and "Uebersetzung: 1st International Collection of Tongue Twisters" (Uebersetzung n.d.). *Omniglot* is "an encyclopedia of writing systems and languages" that provides users with advice about L2 learning and teaches phrases and tongue-twisters in over 890+ languages. *Uebersetzung* has the world's largest collection of tongue-twisters, with 3,660 entries from approximately 118 languages. Consider the following tongue-twisters from *Uebersetzung* on the pronunciation of /s/ in Spanish:

Si la sierva que te sirve, no te sirve como sierva, de que sirve que te sirvas de una sierva que no sirve, La sucesión sucesiva de sucesos sucede sucesivamente con la sucesión del tiempo, Si Sansón no sazona su salsa con sal, le sale sosa, le sale sosa su salsa a Sansón si la sazona sin sal.

(*Uebersetzung* n.d.)

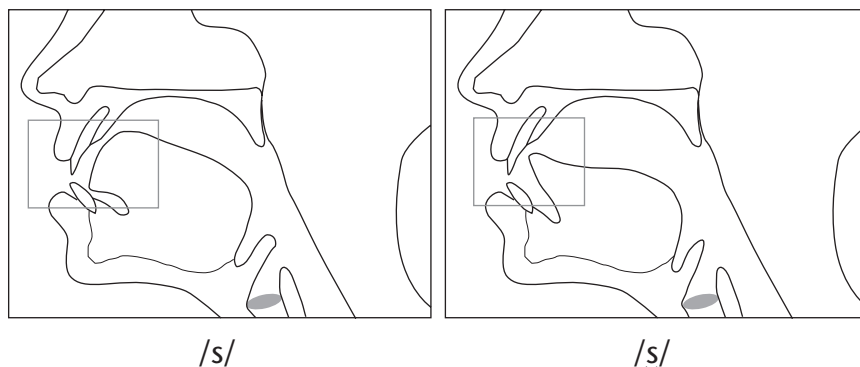


FIGURE 4.9 Facial diagram comparing lamino-alveolar to apical-alveolar /s/ in Spanish.

Source: adapted from Hammond (2001, 212).

4.5.3 The voiceless velar fricative /x/

The phoneme /x/ is a voiceless velar fricative with several different articulations, depending on the dialect. It is pronounced by placing the back of the tongue near the **velum** without blocking the airstream. As air passes through the canal, audible friction is produced. In Spain, it is articulated as [χ], a voiceless uvular fricative where the back of the tongue approaches the uvula (Hualde 2005, 156). Air is pushed through the canal with much greater force and friction in [χ] when compared to [x].

Spanish [x] is represented orthographically as <j> (e.g., *jota*, *garaje*, *jalapeño*) and as <g> before <e> and <i> (e.g., *girar*, *ginebra*, and *gente*). In Mexico, the grapheme <x> may be articulated as [x] or [h] (e.g., in *México*, *Oaxaca*, *Xalapa*, and *Tlaxiaco*). Mexican words that are derived from Nahuatl (e.g., *Xochimilco*, *Xochicalco*) have variant pronunciations where <x> is pronounced as [s], while in other words, <x> is usually pronounced as [ks] (e.g., *exacto* [ek.'sak.ɬo], *excelente* [ek.se.'lep.ɬe]). Table 4.7 lists the distribution of /x/ and its dialectal variants [h, χ, s], along with a formal phonological rule that generates each allophone.

Dalbor (1980, 100) notes that L2 speakers may have difficulty pronouncing [x], especially if this sound does not exist in their L1. He notes, however, that some may choose to articulate this sound as [h] since it is an acceptable substitute in many dialects of Spanish. Students can be told that [h] can be used instead of [x]; however, teachers should be aware that students might overgeneralize this to Spanish words that are written with mute <h>.

Classroom activities should consist of recognition exercises where students listen to a list of words and are required to indicate whether they are written with a <g>, <j>, or <x> (in the case of Mexican toponyms). This could be followed by practice dictation exercises contrasting the graphemes <g>, <j>, and <x> when they

TABLE 4.7 Distribution of /x/

Allophone	Description	Environment	Examples		
			Word	Phonemic transcription	Phonetic transcription
/x/ [x]	[-voiced] alveolar	everywhere	<i>gente</i>	/xente/	['xep.ɬe]
[h]*	fricative		<i>jamón</i>	/xamon/	[ha.'mon]
[χ]*	[-voiced] glottal		<i>jota</i>	/xota/	['χo.ɬa]
[s]†	fricative		<i>Xochimilco</i>	/xotʃimilko/	[so.tʃi.'mil.ko]
	[-voiced] uvular				
	fricative				
	[-voiced] alveolar				
	fricative				

Notes

* denotes acceptable dialectal differences;

† denotes words in Mexican Spanish, usually toponyms, derived from Nahuatl.

are pronounced as [x]. The sample exercise in Table 4.8 helps students make appropriate mental representations of this sound in relation to corresponding graphemes.

Other activities for teaching /x/ may include facial diagrams for teaching sounds that do not exist in the learner’s L1; for example, the diagrams in Figure 4.10 can be used to compare and contrast the allophones of /x/.

For advanced classes in Spanish phonetics and phonology, students are frequently required to recognize dialectal differences in the pronunciation of specific sounds, for example, how [h, x, χ] differ acoustically. When the goal is to have students simply recognize different pronunciations of /x/, teachers can engage students in auditory discrimination exercises that require them to circle the word that best represents the teacher’s pronunciation.⁷ Consider the exercise in Table 4.9.

TABLE 4.8 Exercise to practice sound-grapheme correspondences for /x/

#	Word	Transcription	Correct response
1.	inteligente	[in̪.te.li.ˈxep̪.te]	g
2.	consejero	[kon.se.ˈxe.ro]	j
3.	mujer	[mu.ˈxer]	j
4.	México	[ˈme.xi.ko]	x
5.	Argentina	[aɾ.xep̪.ˈt̪i.na]	g

TABLE 4.9 Auditory discrimination exercise to practice dialectal variation of /x/

#	Palabra	Pronunciación	Respuesta
1.	gente	[ˈχep̪.te]	[h] [x] (χ)
2.	jugo	[ˈhu.χo]	(h) [x] [χ]
3.	girar	[xi.ˈrar]	[h] (x) [χ]
4.	ginebra	[xi.ˈne.βra]	[h] (x) [χ]
5.	álgebra	[ˈal.he.βra]	(h) [x] [χ]
6.	jota	[ˈχo.ta]	[h] [x] (χ)

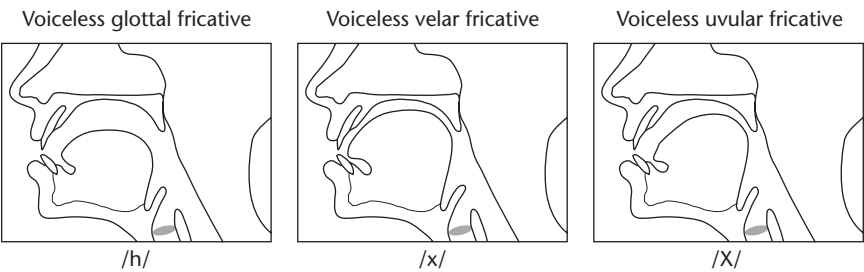


FIGURE 4.10 Facial diagrams comparing and contrasting Spanish [h, x, χ].

Source: adapted from UIRF (n.d.).

Depending on the goals and the level of the course, the above exercise can be modified to include additional dialectal differences in the articulation of /x/, for example, the Chilean voiceless **palatal** fricative [ç] before the front vowels [e, i] in words such as *gente* [ˈçeɲ̞.te] or *girar* [çi.ˈrar].

4.6 Conclusions

The purpose of this chapter was to provide an overview of research studies examining the acquisition of fricatives by L2 learners of Spanish, as well as the practical application of these theories to classroom methodologies. In this chapter, we discussed research that identifies variables that account for, and perhaps explain, underlying processes in the acquisition of L2 pronunciation. In addition, we learned that study abroad is important for developing student awareness of dialectal differences; however, this does not guarantee that learners will use these sounds when communicating with native speakers. Although more research is needed, using technology is promising for helping students develop their awareness, perception, and articulation of Spanish fricatives, and also aids teachers in the development of pronunciation lessons and practice exercises. In spite of the many advantages of formal phonetic instruction, learners still fall short in developing native or near-native pronunciation. Consequently, many educators tend to view this type of instruction as the least appealing and overly form-focused, in addition to not being concordant with “communicative, meaning-focused methodologies” (Arteaga 2000; Kissling 2013).

As researchers and teachers, we need to re-examine the role of pronunciation instruction in language curricula. Detailed phonetic instruction is generally offered at advanced levels, while it is sorely lacking in introductory courses. Perhaps waiting so long to provide more in-depth pronunciation instruction is beyond the “optimal stage in the L2 acquisition process or an optimal moment in the L2/FL curriculum” (Kissling 2013, 8). Kissling notes that if instruction leads to short-term improvements in pronunciation, then perhaps it “is best provided early in the curriculum and then revisited in advanced courses” (Kissling 2013, 9). If helping students develop proper L2 pronunciation is our goal, it is important to provide learners with quality pronunciation lessons at all levels of instruction.

Notes

- 1 Although many introductory textbooks in Spanish, for example *Puntos de partida* (Dorwick et al. 2017), *Ariba* (Zayas-Bazan, Bacon, and Nibert 2012), and *Dicho y hecho* (Potowski, Sobral, and Dawson 2015), and advanced Spanish phonetics and phonology books classify [β, ð, γ] as fricatives, we believe the sounds are more appropriately classified as approximants, which are covered in detail in Chapter 3 of this volume. Subsequently, [β, ð, γ] will not be included in this chapter.
- 2 Spanish /j/ is better described as a voiced palatal approximant and will not be discussed here.
- 3 Several respondents listed intervocalic “fricatives” *b*, *d*, and *g* as problematic for L2 learners of Spanish; however, as stated here and elsewhere, these sounds are more aptly described as approximants.

- 4 Later in this chapter, I argue that [h] can be an appropriate substitute for /x/, which, according to CA, should not be difficult for students to acquire, regardless of their L1. On the other hand, [h] may also occur as an allophone of /s/ in dialects in which /s/ is aspirated in coda position, for example, *¿Cómo estás?* ['ko.mɐh.'t̪ah]. Research on L2 phonological acquisition shows that allophonic differences are more difficult to acquire than are phonemic differences, so an a priori hypothesis would predict that [h] as an allophone of /s/ should present higher levels of difficulty for L2 learners of Spanish. For situations in which students are planning to travel to areas where /s/-aspiration dialects are spoken, it might behoove them to learn this particular dialectal pronunciation variant as well.
- 5 There are other examples not discussed here where English <s> may influence L2 Spanish pronunciation; for example, intervocalic <s> in polysyllabic words may be pronounced as [ʒ] in English (e.g., *measure* ['mɛ.ʒə] and *vision* ['vi.ʒən]), which may lead students to pronounce the Spanish word *visión* as [vi.'ʒjo:n] and not [bi.'sjon].
- 6 The student's pronunciation of /s/ in ['ro.sa] in Figure 4.8 was voiceless with a duration of approximately 143 milliseconds (ms). We subsequently compared this to a recording of a native speaker's pronunciation of the same word whose intervocalic /s/ measured 73 ms, or approximately 48.9 percent shorter in duration in comparison to our English-speaking student's pronunciation of this sound.
- 7 It is recommended that the teacher practice pronouncing this exercise prior to administering it in class. Many aspirated forms of /x/ may be pronounced in conjunction with fricative [x], so the final outcome is really [xh], a coarticulated (see **coarticulation**) sound. This may make it difficult for students to differentiate among these sounds if coarticulated. Therefore, care must be taken to avoid coarticulation.

References

- Ager, Simon. n.d. "Tongue Twisters in Spanish." *Omniglot: The Encyclopedia of Writing Systems & Languages*. Retrieved June 14, 2017 from <http://omniglot.com/language/tonguetwisters/spanish.htm#spanish>.
- Alvord, S. M. and D. E. Christiansen. 2012. "Factors Influencing the Acquisition of Spanish Voiced Stop Spirantization during an Extended Stay Abroad." *Studies in Hispanic and Lusophone Linguistics* 5 (2): 239–276.
- Arteaga, D. L. 2000. "Articulatory Phonetics in the First-Year Spanish Classroom." *The Modern Language Journal* 84 (3): 339–354.
- Azevedo, M. M. 2008. *Introducción a la lingüística española*, 3rd edn. Upper Saddle River, NJ: Prentice Hall.
- Bajuniemi, A. 2013. "Teaching Intervention on the Pronunciation of Spanish Intervocalic /d/." In *Selected Proceedings of the 15th Hispanic Linguistics Symposium*, eds. C. Howe, S. E. Blackwell, and M. Lubbers Quesada, 39–50. Somerville, MA: Cascadia Proceedings Project.
- Boersma, P. and D. Weenink. 2015. *Praat: Doing Phonetics by Computer*. www.fon.hum.uva.nl/praat/.
- Bowen, D. J. and R. P. Stockwell. 1960. *Patterns of Spanish Pronunciation a Drillbook*. Chicago: University of Chicago Press.
- Castino, J. M. 1992. "Markedness as a Predictor of Difficulty in the Second Language Acquisition of Spanish Phonology." PhD diss., University of Pittsburgh. <http://search.proquest.com.ezproxy.uta.edu/docview/303961912/abstract>.
- Dalbor, J. B. 1980. *Spanish Pronunciation Theory and Practice*, 2nd edn. Fort Worth, TX: Harcourt College.
- Díaz-Campos, M. 2004. "Context of Learning in the Acquisition of Spanish Second Language Phonology." *Studies in Second Language Acquisition* 26 (2): 249–273.

- Dorwick, T., A. M. Pérez-Girones, A. Becher, C. Isabelli, and A. R. Elliott. 2017. *Puntos de partida: An Invitation to Spanish LAE*, 10th edn. San Francisco, CA: McGraw-Hill Education.
- Elliott, A. R. 1995a. "Field Independence/Dependence, Hemispheric Specialization, and Attitude in Relation to Pronunciation Accuracy in Spanish as a Foreign Language." *The Modern Language Journal* 79 (3): 356–371. doi:10.2307/329351.
- Elliott, A. R. 1995b. "Foreign Language Phonology: Field Independence, Attitude, and the Success of Formal Instruction in Spanish Pronunciation." *The Modern Language Journal* 79 (4): 530–542. doi:10.2307/330005.
- Elliott, A. R. 1997. "On the Teaching and Acquisition of Pronunciation within a Communicative Approach." *Hispania* 80 (1): 95–108. doi:10.2307/345983.
- Elliott, A. R. 2003. "Staking out the Territory at the Turn of the Century: Integrating Phonological Theory, Research, and the Effect of Formal Instruction on Pronunciation in the Acquisition of Spanish as a Second Language." In *Spanish Second Language Acquisition: State of the Science*, eds. B. A. Lafford and M. Rafael Salaberry, 19–46. Washington, DC: Georgetown University Press.
- Face, T. L. and M. R. Menke. 2009. "Acquisition of the Spanish Voiced Spirants by Second Language Learners." In *Selected Proceedings of the 11th Hispanic Linguistics Symposium*, eds. J. Collentine, M. García, B. Lafford, and F. Marcos Marín, 39–52. Somerville, MA: Cascadia Proceedings Project.
- Geeslin, K. L. and A. Gudmestad. 2008. "The Acquisition of Variation in Second-Language Spanish: An Agenda for Integrating Studies of the L2 Sound System." *Journal of Applied Linguistics and Professional Practice* 5 (2): 137–157. doi:10.1558/japl.v5i2.137.
- George, A. 2014. "Study Abroad in Central Spain: The Development of Regional Phonological Features." *Foreign Language Annals* 47 (1): 97–114. doi:10.1111/flan.12065.
- González-Bueno, M. 1995. "Adquisición de los alófonos fricativos de las oclusivas sonoras españolas por aprendices de español como segunda lengua." *Estudios de Lingüística Aplicada* 21: 64–79. <http://ela.cele.unam.mx/index.php/ela/article/view/278>.
- González-Bueno, M. 1997. "The Effect of Formal Instruction on the Acquisition of Spanish Phonology." In *Contemporary Perspectives on the Acquisition of Spanish*, eds. W. R. Glass and A. T. Pérez-Leroux, 57–75. Somerville, MA: Cascadia Press.
- Hammond, R. M. 2001. *The Sounds of Spanish: Analysis and Application*. Somerville, MA: Cascadia Press.
- Hualde, J. I. 2005. *The Sounds of Spanish*. Cambridge: Cambridge University Press.
- Hualde, J. I., A. Olarrea, A. M. Escobar, and C. E. Travis. 2010. *Introducción a la lingüística hispánica*, 2nd edn. Cambridge: Cambridge University Press.
- Kissling, E. M. 2013. "Teaching Pronunciation: Is Explicit Phonetics Instruction Beneficial for FL Learners?" *Latin American, Latino and Iberian Studies Faculty Publications*. Paper 8. Retrieved March 30, 2016 from <http://scholarship.richmond.edu/lalis-faculty-publications/8>.
- Knouse, S. M. 2012. "The Acquisition of Dialectal Phonemes in a Study Abroad Context: The Case of the Castilian Theta." *Foreign Language Annals* 45 (4): 512–542. doi:10.1111/j.1944-9720.2013.12003.x.
- Lado, R. 1957. *Linguistics Across Cultures: Applied Linguistics for Language Teachers*. Ann Arbor, MI: University of Michigan Press.
- Ladefoged, P. and I. Maddieson. 1996. *The Sounds of the World's Languages*, 1st edn. Oxford: Wiley-Blackwell.
- Lord, G. 2005. "(How) Can We Teach Foreign Language Pronunciation? On the Effects of a Spanish Phonetics Course." *Hispania* 88 (3): 557–567. doi:10.2307/20063159.
- Lord, G. 2008. "Podcasting Communities and Second Language Pronunciation." *Foreign Language Annals* 41 (2): 364–379.
- Lord, G. 2010. "The Combined Effects of Immersion and Instruction on Second Language Pronunciation." *Foreign Language Annals* 43 (3): 488–503.

- McBride, K. 2015. "Which Features of Spanish Learners' Pronunciation Most Impact Listener Evaluations?" *Hispania* 98 (1): 14–30. doi:10.1353/hpn.2015.0001.
- MacDonald, D., G. Yule, and M. Powers. 1994. "Attempts to Improve English L2 Pronunciation: The Variable Effects of Instruction." *Language Learning* 44 (1): 75–100.
- Martinet, A. 1980. "De divers types de consonnes continues in Homenaje a Ambrosio Rabanales." *Boletín de Filología Santiago de Chile* 31: 435–442.
- Martínez Celdrán, E. 2004. "Problems in the Classification of Approximants." *Journal of the International Phonetic Association* 34 (2): 201–210. doi:10.1017/S0025100304001732.
- Potowski, K., S. Sobral, and L. M. Dawson. 2015. *Dicho y hecho: Beginning Spanish*, 10th edn. Hoboken, NJ: Wiley.
- Ringer-Hilfinger, K. 2012. "Learner Acquisition of Dialect Variation in a Study Abroad Context: The Case of the Spanish [θ]." *Foreign Language Annals* 45 (3): 430–446.
- Schmidt, L. B. 2014. "Contextual Variation in L2 Spanish: Voicing Assimilation in Advanced Learner Speech." *Studies in Hispanic and Lusophone Linguistics* 7 (1): 79–113.
- Schmidt, L. B. and E. W. Willis. 2011. "Systematic Investigation of Voicing Assimilation of Spanish /s/ in Mexico City. In *Selected Proceedings of the Fifth Conference on Laboratory Approached to Romance Phonology*, ed. S. Alvord, 1–20. Somerville, MA: Cascadia Proceedings Project.
- Stockwell, R. P. and J. D. Bowen. 1965. *The Sounds of English and Spanish*. Contrastive Structure Series. Chicago, IL: University of Chicago Press.
- Teschner, R. V. 2000. *Camino oral: fonética, fonología y práctica de los sonidos del español + Student Audio CD Program*, 2nd edn. Boston, MA: McGraw-Hill Education.
- Thomas, J. M. C., L. Bouquiaux, and F. Cloarec-Heiss. 1986. *Iniciación a la fonética: fonética articulatoria y fonética distintiva*. Madrid: Gredos.
- Uebersetzung. n.d. "1st International Collection of Tongue Twisters." Retrieved June 14, 2017 from www.uebersetzung.at/twister/.
- UCLA Phonetics Lab Software. 2016. Retrieved March 29, 2016 from www.linguistics.ucla.edu/faciliti/sales/software.htm.
- UIRF (University of Iowa Research Foundation) (n.d.). *Phonetics: The Sounds of Spanish. Fonética: Los Sonidos del Español*. Retrieved March 23, 2016, from <https://soundsofspeech.uiowa.edu/main/spanish>.
- Wardhaugh, R. 1970. "The Contrastive Analysis Hypothesis." *TESOL Quarterly* 4 (2): 123–130.
- Zampini, M. L. 1994. "The Role of Native Language Transfer and Task Formality in the Acquisition of Spanish Spirantization." *Hispania* 77 (3): 470–481. doi:10.2307/344974.
- Zayas-Bazan, E. J., S. Bacon, and H. J. Nibert. 2012. *Arriba: comunicación y cultura*, 6th edn. Boston, MA: Prentice Hall.